

Workshop Activities

Activity 1 – Creating Column Charts

Consider the Cheetah Sports regional sales data included in the *CheetahRegion* data file shown below.

	A	B	C	D
1	Year	Eastern Sales	Western Sales	Total Sales (in \$ millions)
2	1	59	28	87
3	2	57	33	90
4	3	68	42	110
5	4	91	54	145
6	5	109	61	170
7	6	96	58	154
8	7	110	67	177
9	8	72	103	175
10	9	63	120	183
11	10	65	130	195

Open the *CheetahRegion* sports sales data file and create a column chart using the following steps:

- Step 1. Select cells A1:B11
- Step 2. Click the Insert tab on the Ribbon.
- Step 3. In the Charts group, click the Insert Column or Bar Chart button
- Step 4. Select Clustered Column

The resulting chart is incorrectly rendered because Excel displays the year as if it was a quantitative variable. To correct this, use the following additional steps:

- Step 5. Right-click on the chart and select Change Chart Type
- Step 6. When the Change Chart Type task pane opens, select the Cluster Column type that plots the single variable Sales with ten monochromatic columns, and click OK

As described in Module 1 - Activity 5, edit the chart to improve its appearance.

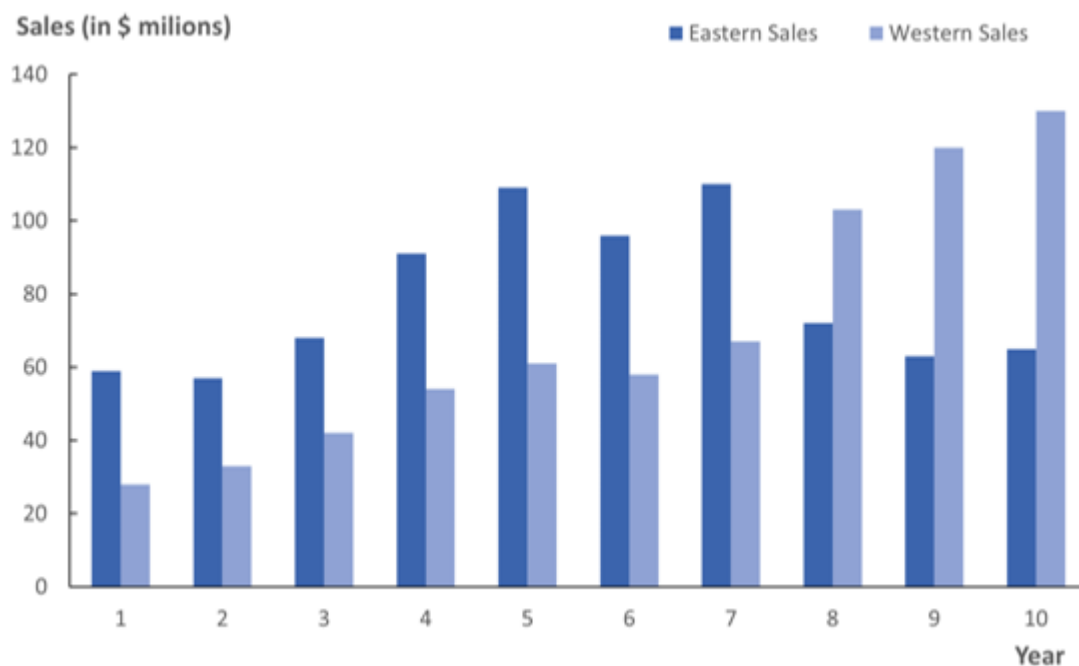
Then add data labels to the columns:

- Step 7. Click on the Chart Elements button and select Data Labels.

Now create a clustered column chart using the same data.

- Step 1. Select cells A1:C11 (do not select cells D1:D11)
- Step 2. Follow Steps 2-6 previously outlined for a column chart, including the change of colour scheme to Monochromatic Palette 1 (using the Chart Styles option)

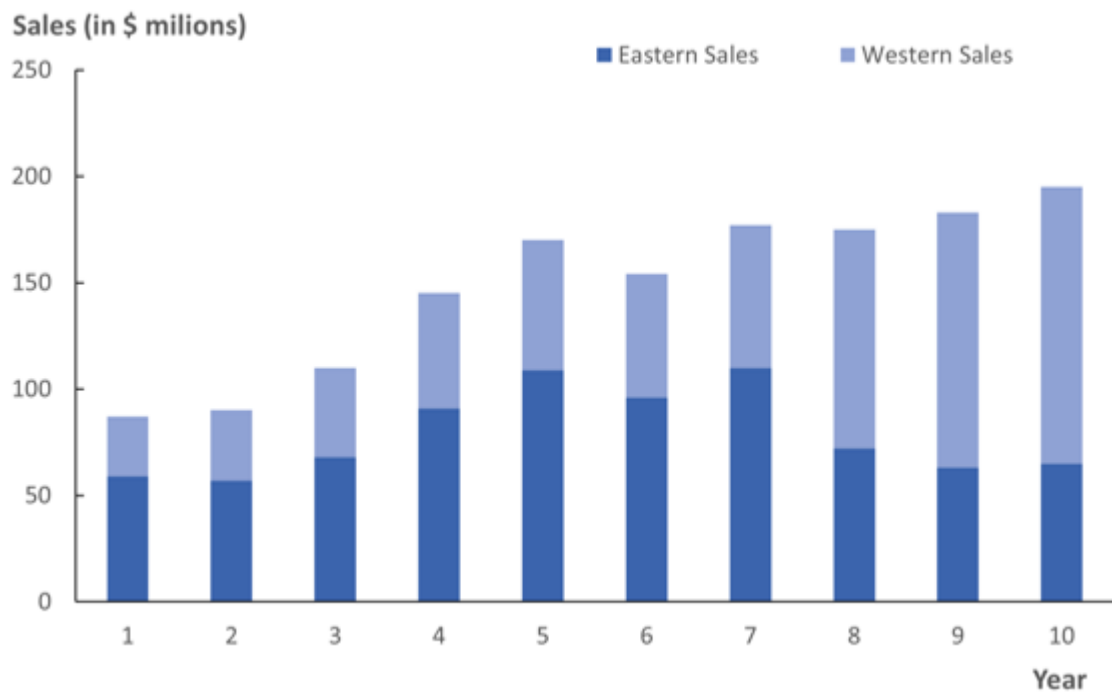
Your clustered chart should now look similar to the following chart:



Now create a stacked column chart using the same *CheetahRegion* sports sales data.

- Step 1. Select cells A1:C11 (do not select cells D1:D11)
- Step 2. Follow Steps 2-6 previously outlined for a column chart, except that in Step 4 click on the Insert Column or Bar Chart in the Charts group and select Stacked Column

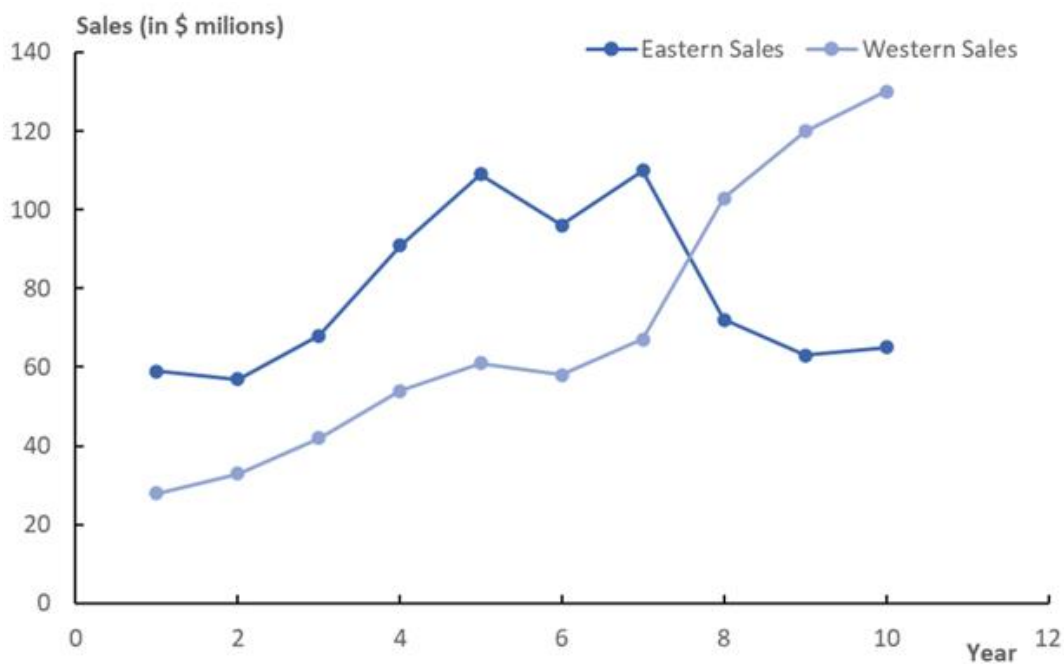
Your stacked column chart should look similar to the following chart:



Consider:

Why does the stacked column chart make the preferred data visualisation choice over the clustered column chart?

Provide two different examples of why an audience would prefer to see the Cheetah Sports regional data visualised with (1) a line chart (see next page) vs. (2) a stacked column chart.



Line Chart showing *CheetahRegion* sports sales data.

Activity 2 – Creating a Line Chart

The next activity uses the following data from the *Cheetah* data file to create a line chart.

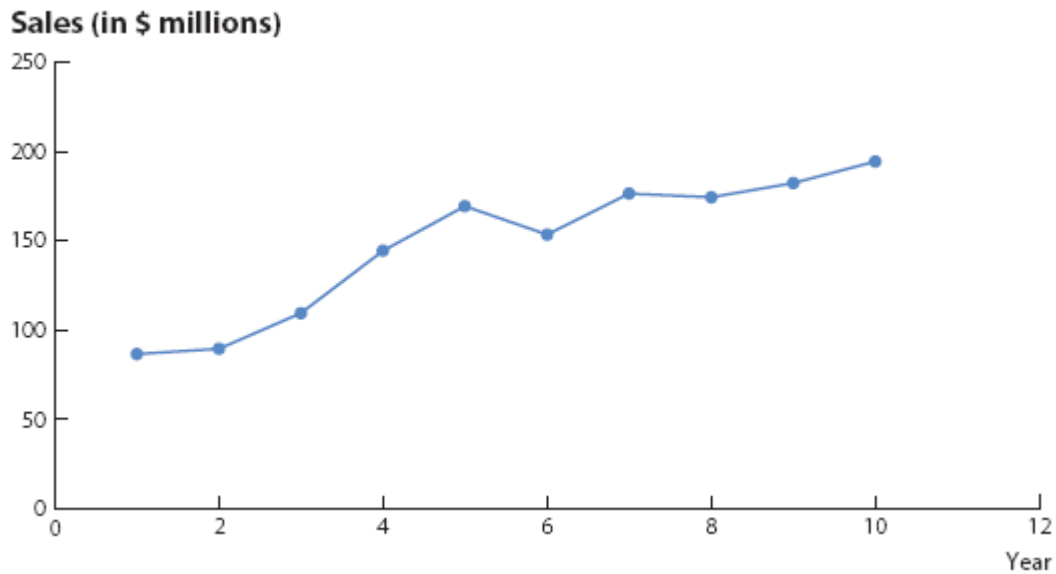
	A	B
1	Year	Sales (in \$ millions)
2	1	87
3	2	90
4	3	110
5	4	145
6	5	170
7	6	154
8	7	177
9	8	175
10	9	183
11	10	195

Open the *Cheetah* data file and complete the following steps:

- Step 1.
 - Select cells A1:B11
- Step 2.
 - Click the Insert tab on the Ribbon.
- Step 3.
 - In the Charts group, click the Insert Scatter (X,Y) or Bubble Chart button
 - Select Scatter with Straight Lines and Markers

Finally, edit the chart to improve its appearance (as described in Module 1 - Activity 5).

Your finished line chart should look similar to the following chart:

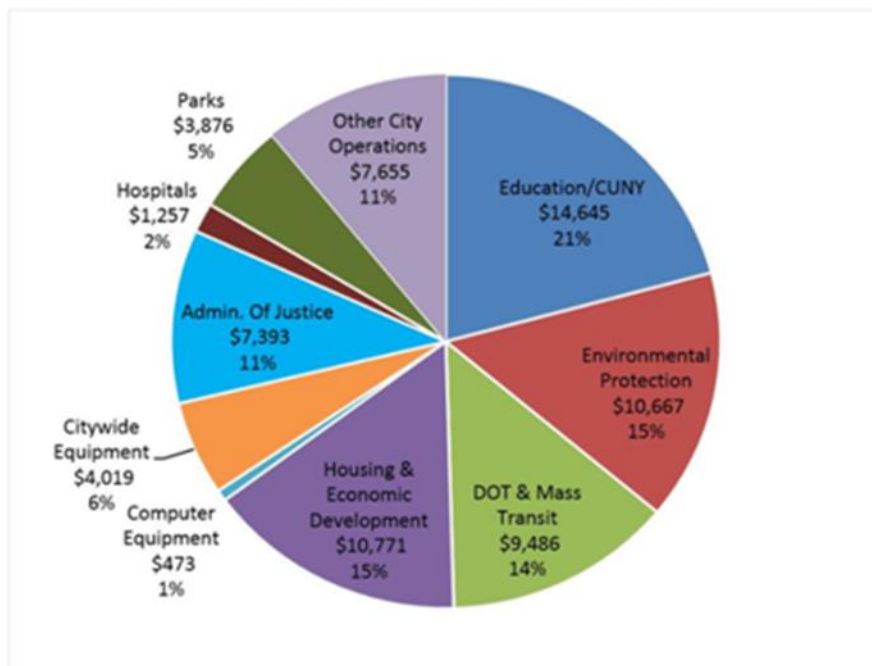


The lines between the points suggests continuity. It makes it easier for the reader/viewer to see and interpret changes that have occurred over time

Activity 3 – Interpreting Charts

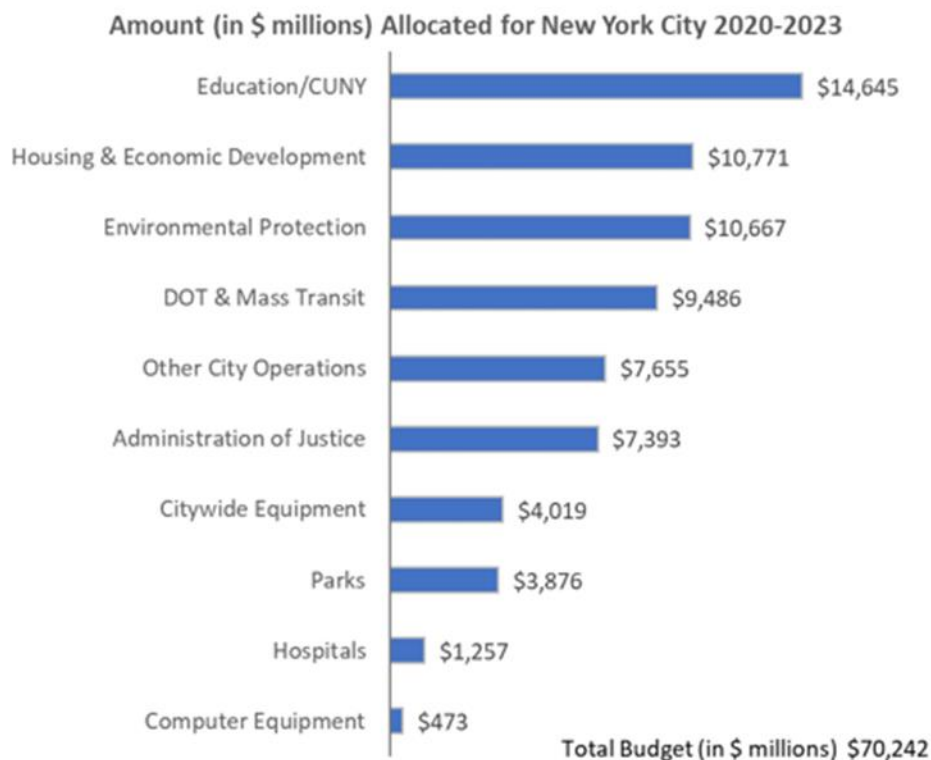
Consider the New York City Budget Allocation case shown in the graphs on this page and the next.

A pie chart showing the NYC budget allocation



Source: NYC Office of Management and Budget, FY 2020 Adopted Capital Commitment Plan, October 2019 Pie Chart figures may not tie due to rounding.

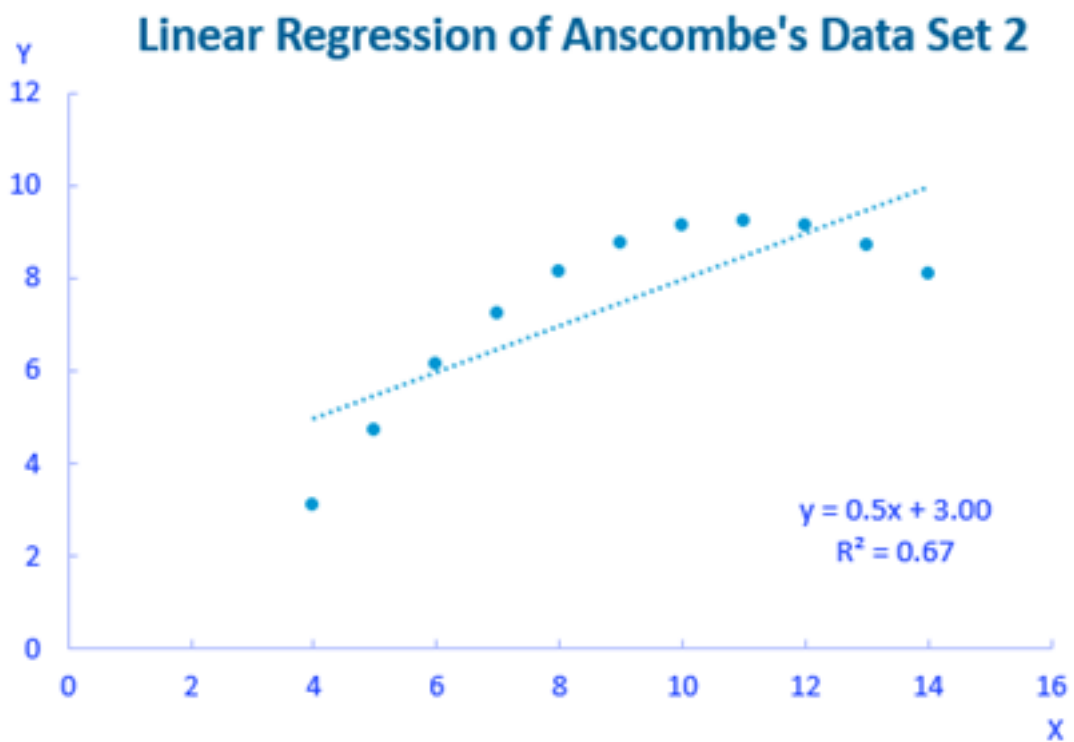
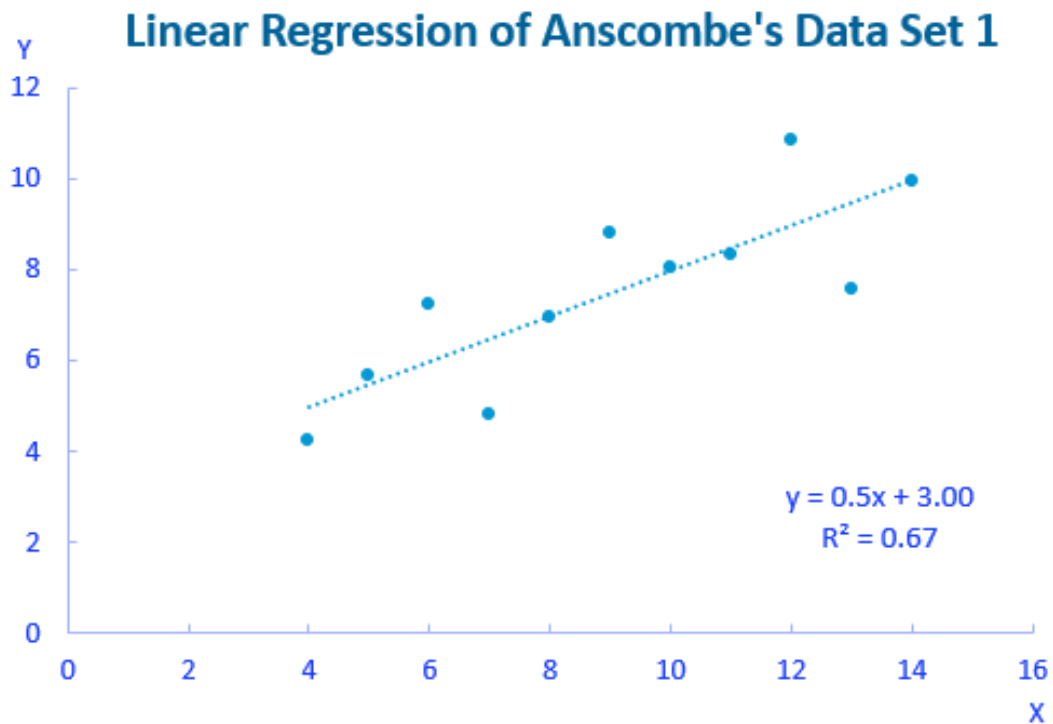
A bar chart showing the NYC budget allocation



1. Give two reasons why the bar chart makes for a better choice than a pie chart to visualise the New York City funds allocation.
2. Why would you choose a bar chart over a column chart to visualise the New York City funds allocation?

Activity 4 – Using Trend Lines

Consider the two scatter charts and related trendline and regression statistics shown below and on the next page, for Anscombe's data sets. The estimated regression equations and related R-squares for both data sets are identical.



Does fitting a line to the data appear to be a wise choice for both data sets? Discuss your answer.